Kai Hori

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EDUCATION

University of California, Davis

Bachelor of Science, ***Aerospace Science and Engineering*** GPA: 3.614/4.0

Bachelor of Science, ***Mechanical Engineering***

Relevant Coursework: Orbital Mechanics, Space Vehicle Design, Stability and Control of Aerospace Vehicles, Automatic Control of Engineering Systems, Mechanics of Materials, Rocket Propulsion, Thermodynamics, Heat Transfer

TECHNICAL SKILLS

CAD & Engineering Software: Onshape, SOLIDWORKS, Fusion 360, ANSYS Fluent, NASTRAN/PATRAN
Programming & Analysis: MATLAB, Simulink, FreeFlyer, Python, C#, Fortran, Unity, HLSL
Manufacturing & Prototyping: 3D Printing, Manual Milling, Lathe, CNC Milling, Drilling, Welding
Certifications: Remote Pilot Certificate with a sUAS rating issued by the FAA (Part 107) August 2026
Languages: English (Fluent), Japanese (Fluent)

RESEARCH EXPERIENCE AND PROJECTS

Mission Operations Intern - ***ispace.inc, S3-1 Lunar Lander Mission Operations*** October 2026 - Present

- Develop standardized Python command-and-telemetry procedures for lunar lander operations, verifying spacecraft responses against expected telemetry.
- Build Grafana telemetry displays for mission-specific scenarios to support real-time operator situational awareness
- Participate in cross-team spacecraft design discussions, representing operations constraints to subsystem engineers**..**

Research Assistant - ***Human/Robotics/Vehicle Integration & Performance Laboratory*** February 2026 - Present

- Design and characterize ground-station RF systems for multi-band weather-satellite reception as part of the Mission Operations and Observation Nexus (MOON) initiative, including a QFH antenna and RF receive chain using LNAs, bandpass filters, and SDRs for 137 MHz Meteor LRPT and 1694 MHz GOES HRIT reception.
- Develop ground-station capabilities for satellite tracking, signal reception, decoding, and image processing to support real-time LEO and GEO satellite operations and mission simulations.

Communications Lead - ***Spacecraft Inspection Mission Design Project, Senior Capstone Class*** April 2026 -June 2026

- Led Communications, C&DH, and Ground Control System design for a heliocentric-orbit satellite inspection mission, developing X-band link and data budgets across multiple mission operating cases, antenna/link-budget sizing, and downlink/TT&C trade studies in coordination with power and payload teams.
- Developed MATLAB and FreeFlyer analyses to evaluate communications performance throughout the mission trajectory and performed modulation/coding trade studies, requirements definition, V&V planning, and FMEA in coordination with spacecraft subsystem leads.

Payload Lead - ***Spacecraft Inspection Mission Redesign Project, Senior Capstone Class*** January 2026 - March 2026

- Led payload and systems design for the redesign of NASA's CPOD CubeSat proximity-operations concept into an autonomous, non-docking spacecraft inspection mission.
- Evaluated LiDAR integration, sensor performance, duty cycle, onboard processing, and spacecraft resource constraints to develop an inspection architecture supporting autonomous RPO and close-range operations.

Research Intern - ***NeuralX*** July 2025 - September 2025

- Designed real-time GPU-parallelized Smoothed Particle Hydrodynamics simulations with 64,000 particles on an Apple M4 chip by writing custom HLSL compute shaders and spatial-neighbor search in C#
- Validated numerical stability of the resulting fluid behavior against expected physical dynamics by iterative process.

RELEVANT EXPERIENCE

Team Member - ***UC Davis Aggie Propulsion Rocketry Lab*** September 2023 - September 2024

- Co-founded a new controls subteam for a liquid rocket engine system and researched system dynamics and control strategies for a propulsive landing system, modeling vehicle behavior to inform control-law development.

OTHER WORK EXPERIENCE

Co-Instructor - ***UC Davis International Center*** June 2024 - June 2026

- Co-taught business fundamentals for agriculture students, translating finance, operations, and marketing concepts into industry-specific applications while serving as a bilingual cultural liaison for the Japanese Agricultural Training Program in partnership with the Japanese Agricultural Exchange Council.